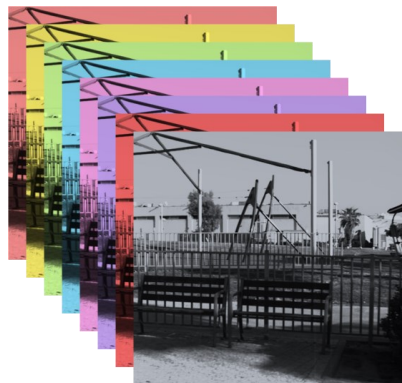
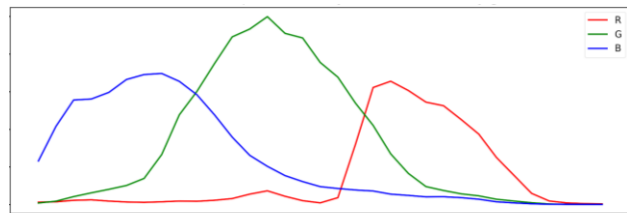




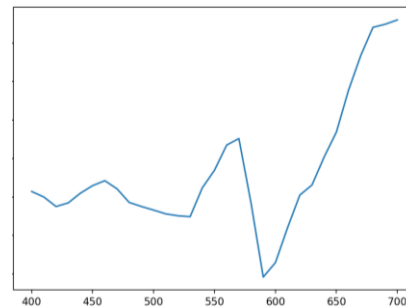
Hyper-spectral images

$$\mathbf{s}_i \in \mathbb{R}^L$$



Estimated camera
sensor sensitivity

$$\mathbf{C} \in \mathbb{R}^{3 \times L}$$



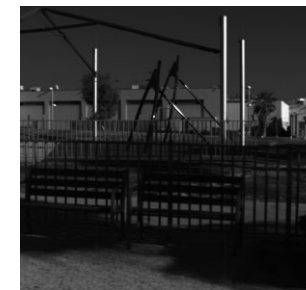
First principal
component of $\mathbf{n}_i$

$$\mathbf{e} \in \mathbb{R}^{1 \times L}$$



Simulated
Camera raw RGB

$$\mathbf{C}\mathbf{s}_i \in \mathbb{R}^3$$

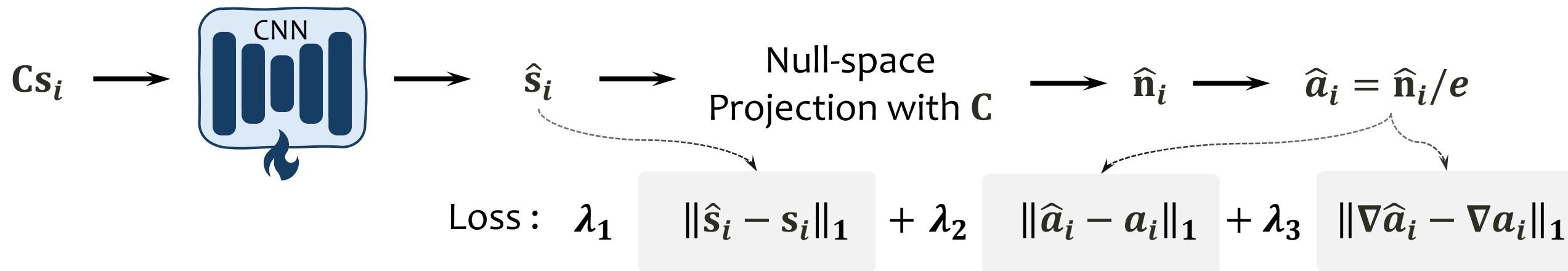


Simulated Camera-
Null coefficient

$$\mathbf{a}_i \in \mathbb{R}^1$$

(a) Estimate $\mathbf{e}$ with calibrated $\mathbf{C}$ and a spectral dataset

(b) Construct virtual training data with (a)



(c) Learning per-pixel camera-null coefficient with CNN network